

Unit 3: JavaScript Input and Output

Introduction

JavaScript provides different methods to **display output** and **receive input** from the user.

These methods help create interactive web pages.

The commonly used Input and Output methods are:

- `alert()`
 - `prompt()`
 - `confirm()`
 - `console.log()`
 - `document.write()`
-

1. `alert()`

Definition

The `alert()` method is used to display a **popup message box** with an **OK** button.

It is mainly used to display notifications or warning messages.

Syntax

```
alert("Message");
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Alert Example</title>
</head>

<body>

<script>

alert("Welcome to JavaScript!");
```

```
</script>
```

```
</body>
```

```
</html>
```

Output

```
+-----+
|       |
| Welcome to JavaScript! |
|       |
|       [ OK ]          |
|       |
+-----+
```

2. prompt()

Definition

The `prompt()` method is used to receive input from the user through a popup dialog box. It displays a text box along with **OK** and **Cancel** buttons.

Syntax

```
prompt("Enter your name");
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Prompt Example</title>
</head>

<body>

<script>

let name = prompt("Enter Your Name");

document.write("Welcome " + name);

</script>

</body>

</html>
```

Output

```
+-----+
| Enter Your Name |
| [ Harikrishnan  ] |
| [ OK ] [ Cancel ] |
+-----+
```

Output:

Welcome Harikrishnan

3. confirm()

Definition

The `confirm()` method displays a popup dialog box with **OK** and **Cancel** buttons.

It returns:

- `true` → OK is clicked.
- `false` → Cancel is clicked.

Syntax

```
confirm("Do you want to continue?");
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Confirm Example</title>
</head>

<body>

<script>

let result = confirm("Do you want to continue?");

document.write(result);

</script>
```

```
</body>
```

```
</html>
```

Output

```
+-----+
| Do you want to continue? |
|       [ OK ]       [ Cancel ] |
+-----+
```

If OK → true

If Cancel → false

4. console.log()

Definition

The console.log() method is used to display messages in the **browser console**.

It is mainly used for debugging and testing JavaScript programs.

Syntax

```
console.log("Message");
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Console Example</title>
</head>

<body>

<script>

console.log("Welcome to JavaScript");

</script>

</body>

</html>
```

Output

Browser Console

Welcome to JavaScript

5. document.write()

Definition

The document.write() method is used to display output directly on the webpage.

Syntax

```
document.write("Message");
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Document Write</title>
</head>

<body>

<script>

document.write("Hello Students!");

</script>

</body>

</html>
```

Output

Hello Students!

Difference Between Input and Output Methods

Method	Purpose
alert()	Displays a popup message
prompt()	Accepts input from the user
confirm()	Displays OK and Cancel buttons
console.log()	Displays output in the browser console
document.write()	Displays output on the webpage

Practical: Create a User Greeting Application

Problem Statement

Create a JavaScript program that asks the user to enter their name using the `prompt()` method and then displays a welcome message using `alert()` and `document.write()`.

Algorithm

Step 1: Create an HTML document.

Step 2: Use the `prompt()` method to get the user's name.

Step 3: Store the entered name in a variable.

Step 4: Display a welcome message using `alert()`.

Step 5: Display the same message on the webpage using `document.write()`.

Step 6: Save the file and run it in a web browser.

Program

```
<!DOCTYPE html>
<html>

<head>

<title>User Greeting Application</title>

</head>

<body>
```

```
<script>

let name = prompt("Enter Your Name");

alert("Welcome " + name + "!");

document.write("<h2>Welcome " + name + "!</h2>");

</script>

</body>

</html>
```

Output

Step 1

```
+-----+
| Enter Your Name |
| [ Harikrishnan  ] |
| [ OK ] [ Cancel ] |
+-----+
```

Step 2

```
+-----+
| Welcome Harikrishnan! |
| [ OK ] |
+-----+
```

Step 3

Welcome Harikrishnan!

HTML Tags Used

Tag	Purpose
<html>	Root element of the webpage
<head>	Contains page information
<title>	Displays the page title

Tag	Purpose
<body>	Displays webpage content
<script>	Contains JavaScript code

JavaScript Methods Used

Method	Purpose
prompt()	Receives input from the user
alert()	Displays a popup message
document.write()	Displays output on the webpage

Learning Outcome

After completing this unit, students will be able to:

- Understand JavaScript input and output methods.
- Use alert(), prompt(), confirm(), console.log(), and document.write().
- Accept input from users.
- Display messages using different output methods.
- Develop simple interactive JavaScript applications.